

Engineering Reboot 2026

Week-02 Engineering Lab Workbook v3.3

Candidate for Lock-In

This workbook is designed as a self-contained execution package. All exercises, file locations, screenshots, notes, journals, and Git commands are specified to minimize ambiguity and context switching.

Day 06 – Conditional Logic

Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible
- ☐ Previous Day Committed And Pushed

Required Source Files

Week-02\Day-06\Exercises\age_classifier.py

Week-02\Day-06\Exercises\temperature_advisor.py

Execution Requirements

All exercises must be executed.

All required test cases must be run.

Expected output must match workbook specifications.

Evidence screenshots must be captured.

Notes File

Week-02\Day-06\Notes\Day-06-Notes.md

Journal File

Week-02\Day-06\Journal\Day-06-Journal.md

Git Command Block

```
git status
git add .
git commit -m "Day 06 Complete"
git push
```

Completion Checklist

- ☐ Exercises Complete
- ☐ Test Cases Executed
- ☐ Screenshots Captured
- ☐ Notes Completed
- ☐ Journal Completed
- ☐ Git Commit Completed
- ☐ Git Push Completed

Day 07 – Boolean Logic

Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible
- ☐ Previous Day Committed And Pushed

Required Source Files

Week-02\Day-07\Exercises\eligibility_checker.py
Week-02\Day-07\Exercises\building_access_control.py

Execution Requirements

All exercises must be executed.
All required test cases must be run.
Expected output must match workbook specifications.
Evidence screenshots must be captured.

Notes File

Week-02\Day-07\Notes\Day-07-Notes.md

Journal File

Week-02\Day-07\Journal\Day-07-Journal.md

Git Command Block

```
git status
git add .
git commit -m "Day 07 Complete"
git push
```

Completion Checklist

- ☐ Exercises Complete
- ☐ Test Cases Executed
- ☐ Screenshots Captured
- ☐ Notes Completed
- ☐ Journal Completed
- ☐ Git Commit Completed
- ☐ Git Push Completed

Day 08 – Loops

Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible
- ☐ Previous Day Committed And Pushed

Required Source Files

Week-02\Day-08\Exercises\counting_program.py
Week-02\Day-08\Exercises\even_number_generator.py
Week-02\Day-08\Exercises\sum_calculator.py

Execution Requirements

All exercises must be executed.
All required test cases must be run.
Expected output must match workbook specifications.
Evidence screenshots must be captured.

Notes File

Week-02\Day-08\Notes\Day-08-Notes.md

Journal File

Week-02\Day-08\Journal\Day-08-Journal.md

Git Command Block

```
git status  
git add .  
git commit -m "Day 08 Complete"  
git push
```

Completion Checklist

- ☐ Exercises Complete
- ☐ Test Cases Executed
- ☐ Screenshots Captured
- ☐ Notes Completed
- ☐ Journal Completed
- ☐ Git Commit Completed
- ☐ Git Push Completed

Day 09 – Loop Practice

Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible
- ☐ Previous Day Committed And Pushed

Required Source Files

Week-02\Day-09\Exercises\square_pattern.py
Week-02\Day-09\Exercises\right_triangle_pattern.py
Week-02\Day-09\Exercises\inverted_triangle_pattern.py
Week-02\Day-09\Exercises\multiplication_table.py

Execution Requirements

- All exercises must be executed.
- All required test cases must be run.
- Expected output must match workbook specifications.
- Evidence screenshots must be captured.

Loop Practice Output Examples

Square Pattern (Input 5)

```
*****  
*****  
*****  
*****  
*****
```

Right Triangle (Input 5)

```
*  
**  
***  
****  
*****
```

Inverted Triangle (Input 5)

```
*****  
****  
***  
**
```

*

Notes File

Week-02\Day-09\Notes\Day-09-Notes.md

Journal File

Week-02\Day-09\Journal\Day-09-Journal.md

Git Command Block

git status

git add .

git commit -m "Day 09 Complete"

git push

Completion Checklist

☐ Exercises Complete

☐ Test Cases Executed

☐ Screenshots Captured

☐ Notes Completed

☐ Journal Completed

☐ Git Commit Completed

☐ Git Push Completed

Day 10 – Mini Project

Pre-Flight Checklist

- ☐ VS Code Open
- ☐ Correct Day Folder Exists
- ☐ Screenshots Folder Exists
- ☐ Notes Folder Exists
- ☐ Journal Folder Exists
- ☐ Git Repository Accessible
- ☐ Previous Day Committed And Pushed

Required Source Files

Week-02\Day-10\Projects\menu_driven_calculator.py

Execution Requirements

- All exercises must be executed.
- All required test cases must be run.
- Expected output must match workbook specifications.
- Evidence screenshots must be captured.

Notes File

Week-02\Day-10\Notes\Day-10-Notes.md

Journal File

Week-02\Day-10\Journal\Day-10-Journal.md

Git Command Block

```
git status
git add .
git commit -m "Day 10 Complete"
git push
```

Completion Checklist

- ☐ Exercises Complete
- ☐ Test Cases Executed
- ☐ Screenshots Captured
- ☐ Notes Completed
- ☐ Journal Completed
- ☐ Git Commit Completed
- ☐ Git Push Completed